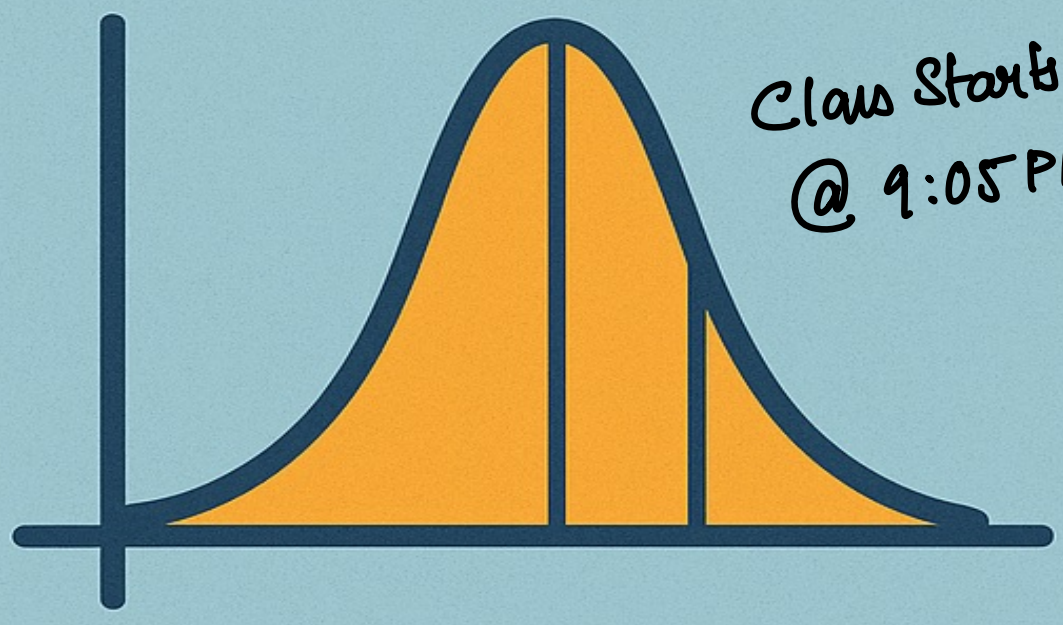


PROBABILITY AND STATISTICS





Lecture 6: Probability distribution - 1

Class starts at 9:05 PM

Agenda

1. Distribution functions
   - → Histogram
   - → PMF
   - → PDF
   - → CDF
2. Empirical Vs Theoritical Probability
3. Binomial distribution + Expectation
4. Bernoulli distribution

Today's class

Random Variable

→ Uncertainity

Example

Rolling a dice

Stars in sky

Rainfall amount

1.0

Continuous

62 68

Types

Discreate RV

for Eg

Rolling a dice

1 2 3

1/2/3.0X

Continuous RV

for Eg

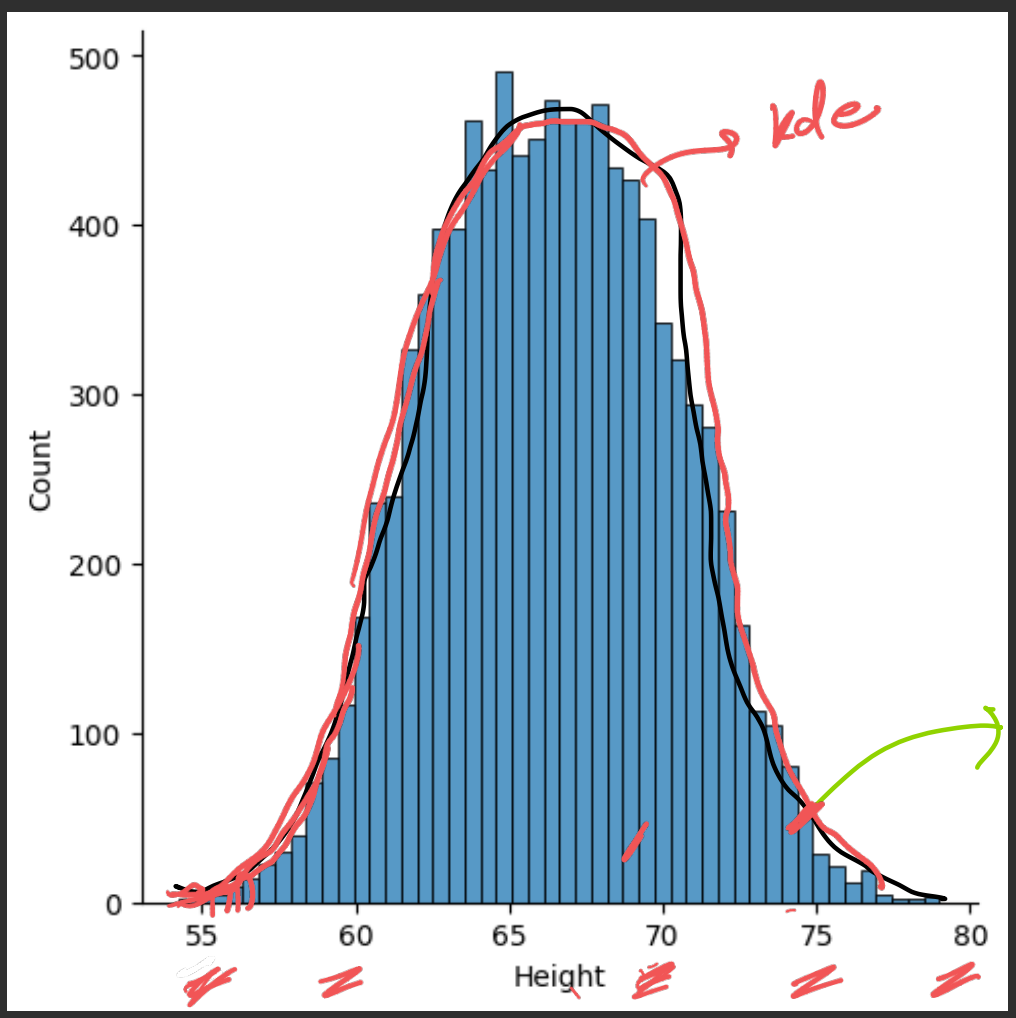
Height of a man

163.4

168.5

1.0

What is distribution?



There are different types of distribution available

Distribution means plotting of data in a histogram

69.2

74.62

74.62

Distribution functions

1) Probability Density function (PDF) (Measuring Rainfall)

↳ Describes the probability density of a continuous RV over its range

2) Probability Mass function (PMF) (Rolling a dice)

↳ Describes the probability of a discrete RV taking on a specific value

3) Cummulative Distribution function (CDF)

↳ gives the probability that a RV is less than or equal to a specified value.

↳ Both CRV/DRV

(pmf) → dist of rolling of a die

62

1, 2, 3, 4, 5, 6

$P(X=1) = 1/6$

$P(X=2) = 1/6$

$P(X=1) = 1/6$

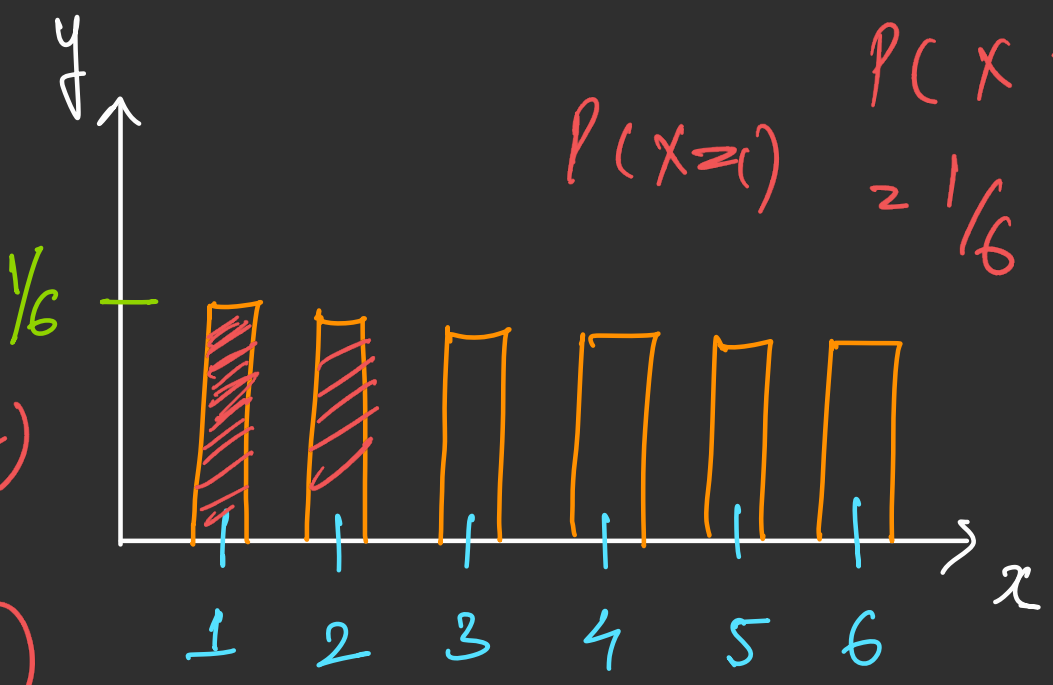
prob

$P(X \leq 5)$

$P(X=1) + P(X=4)$

PDF

$P(X=5)$



$P(X \leq 5)$

$P(X \leq 4)$ = 4 coP

PMF → continuous random variable

$P(X=1)$, $P(X=2)$

$P(X=5)$ → PMF

$P(70 < X < 75)$ → PDF

$P(X = 10) \rightarrow$ PMF (discrete)

$P(X \geq 10) \rightarrow$ CDF

$1 - P(X \leq 10)$

$P(X > 10)$

$P(X \leq 10)$

$1 - P(X \leq 10)$

$P(X \leq 3) \rightarrow$ cdff

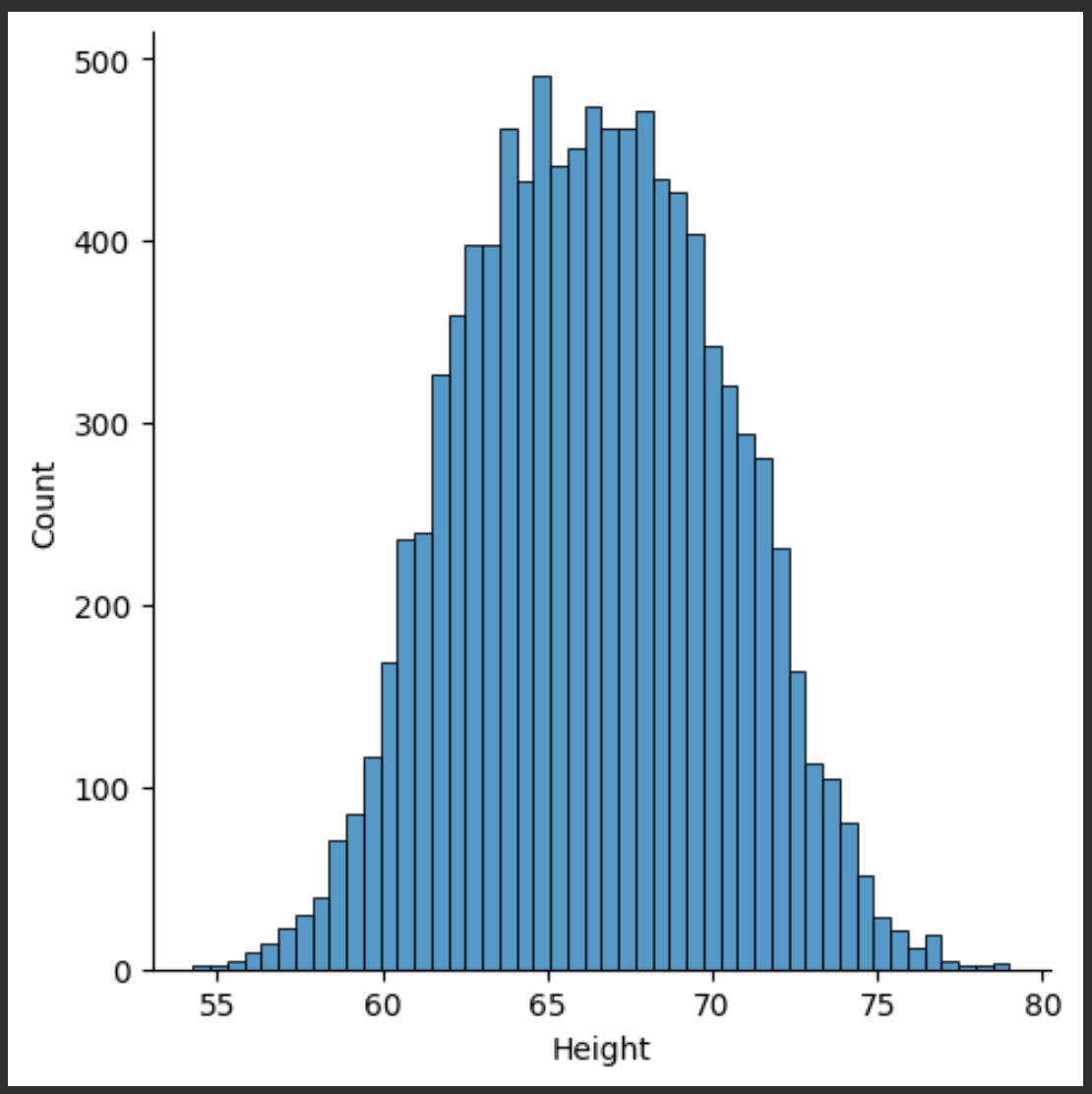
$P(50 < X < 100) \rightarrow$ PDF (Continuous)

Prob Heads $= 0.5$

$1 - 0.5$

$= 0.5$

Histogram



Probability Mass Function (PMF)

Rolling a dice

$X = \{$ o/p of the dice $\}$

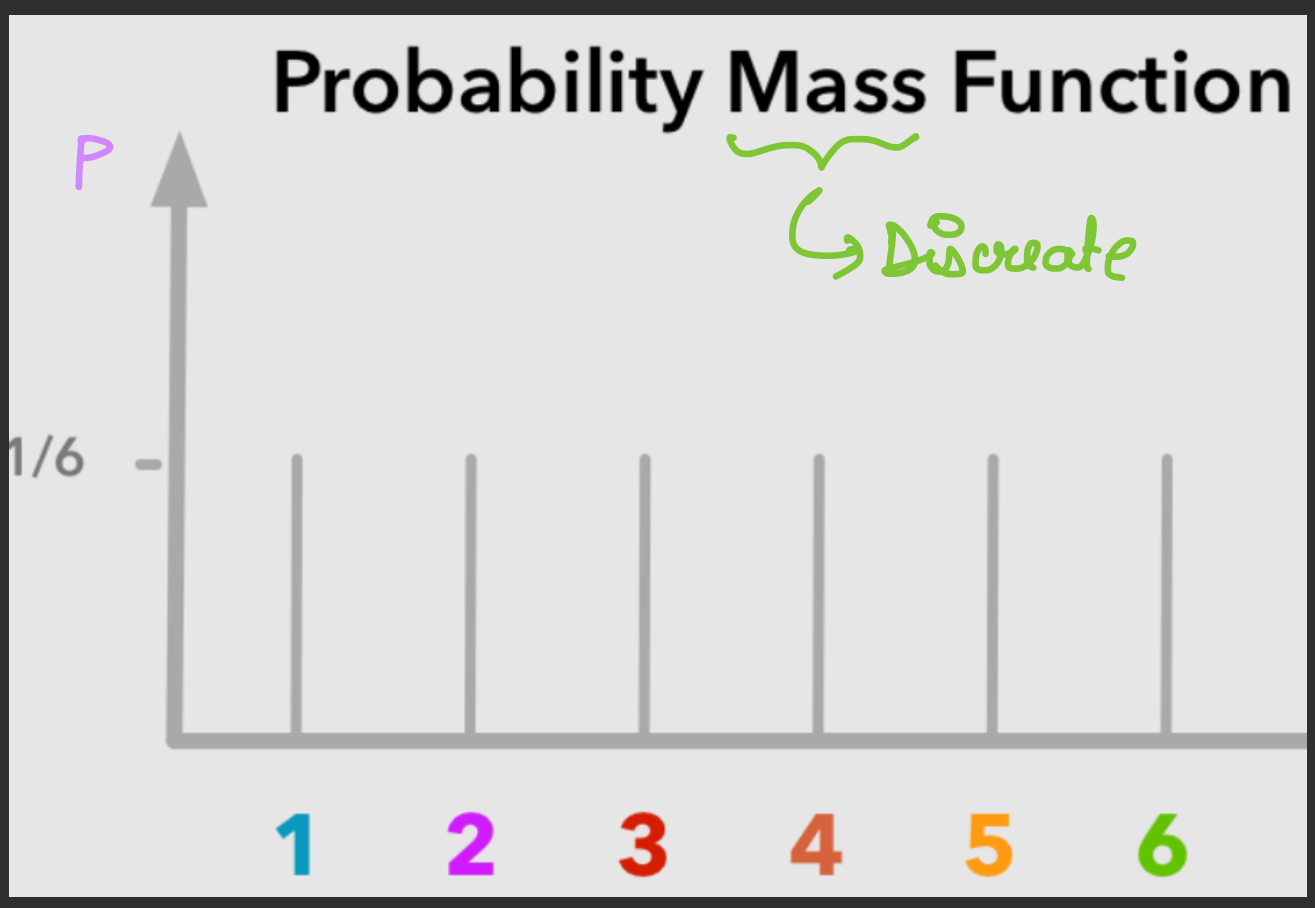
↳ DRV

$P(X=2) = 1/6$

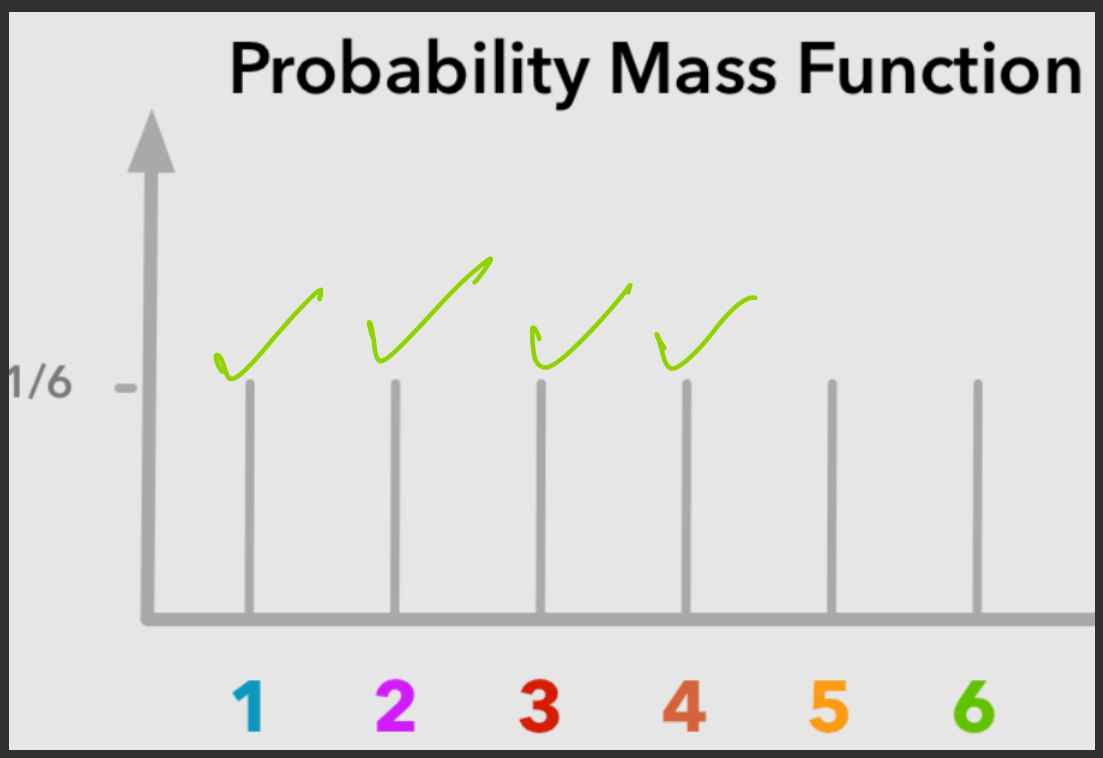
$P(X=2.5) = ?$

Not possible

PMF ⇒ $P(X=2) = 1/6$ This value is the PMF for this case



↳ Discreate



CDF

$P(X \leq 4)$

$P[X=1] + P[X=2] + P[X=3] + P[X=4]$

$\frac{1}{6} + \frac{1}{6} + \frac{1}{6} + \frac{1}{6}$

$\frac{4}{6} = \frac{2}{3}$

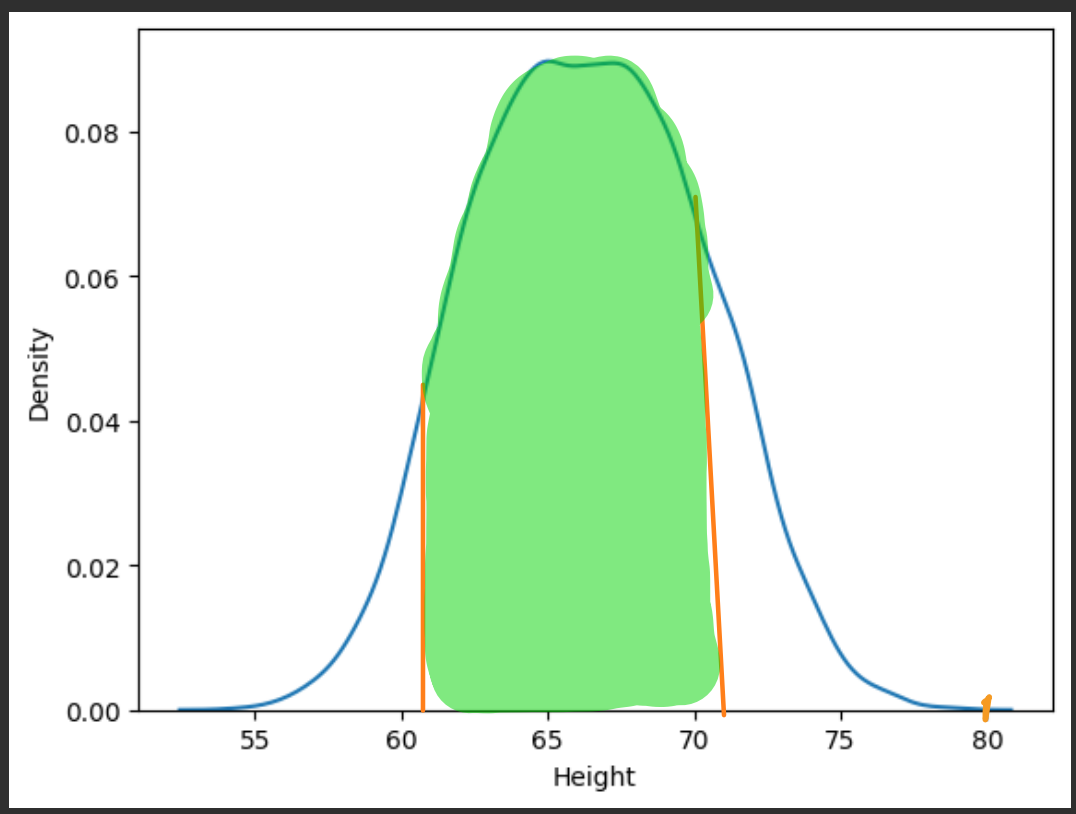
Probability Density function (PDF)

↳ Continuous RV

Height → continuous data

"PDF is used to calculate the range of the prob"

62.5 62.6



P(X > 60 and X < 70)

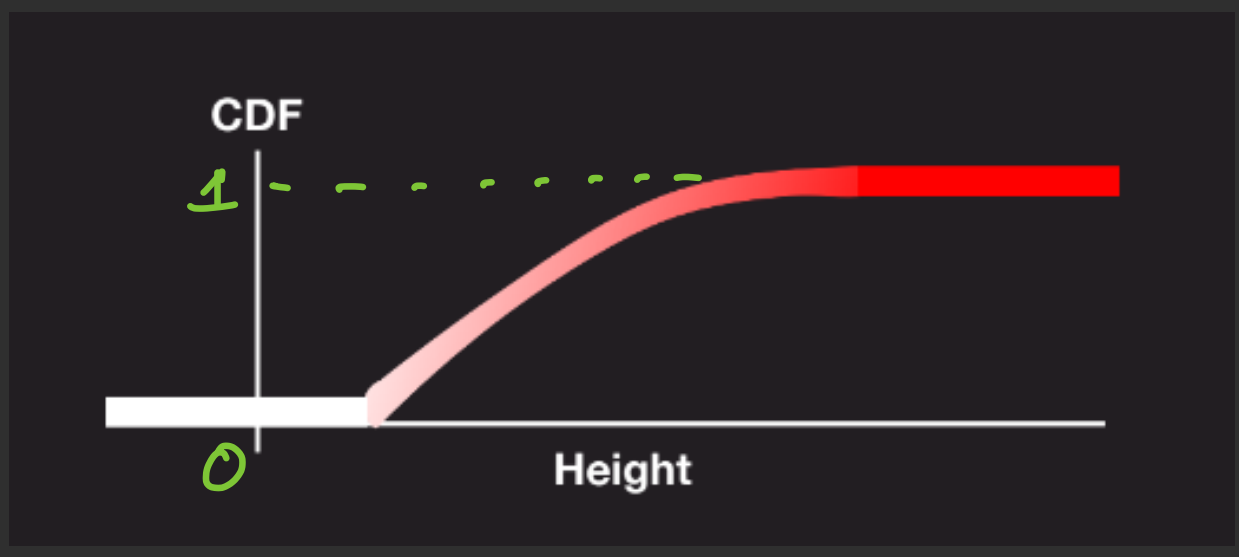
This prob is the PDF

Cumulative Distribution Function (CDF)

1. Can be used for both DRV & CRV
2. CDF ranges from $[0, 1]$

$a = [1, 2, 3, 5]$

- 1
- 1+2 = 3
- 3+3 = 6
- 6+5 = 11



$P(x \leq 70)$

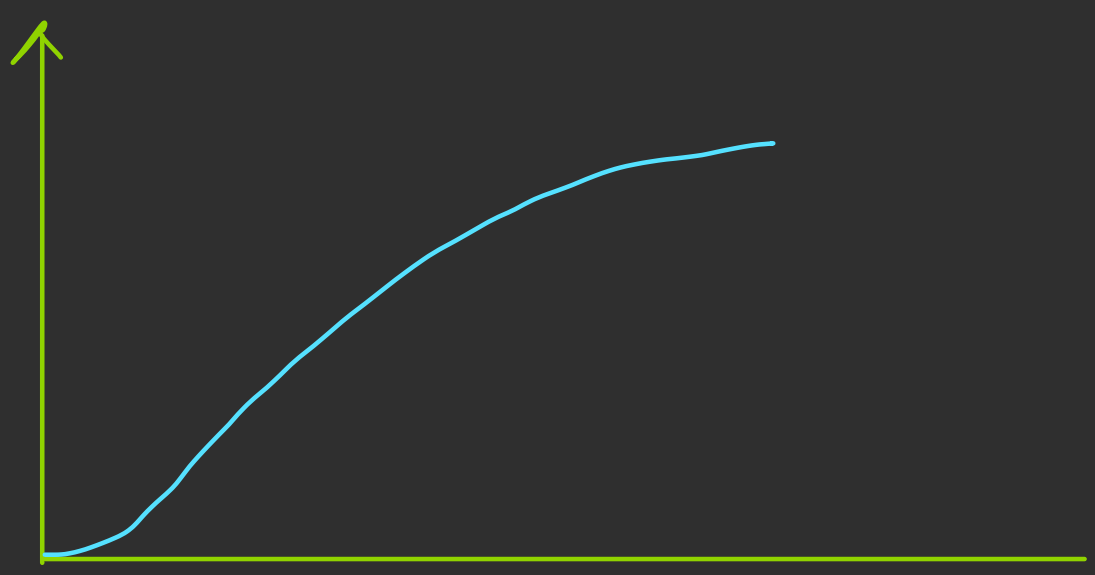
This value will give the CDF

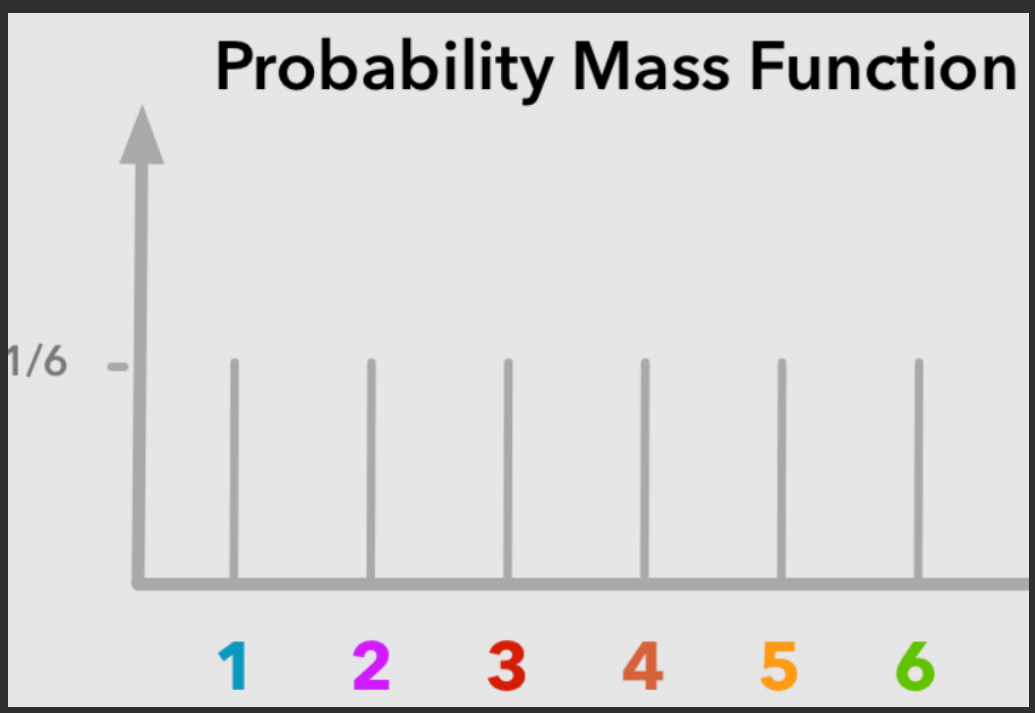
$$P[X=1] = 1/6$$

$$P[X \leq 2] = 2/6$$

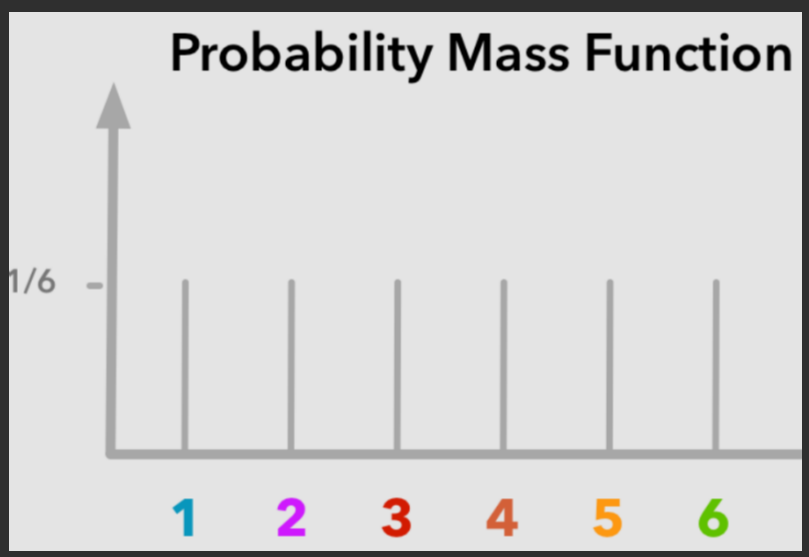
$$P[X \leq 3] = 3/6$$

$$\vdots$$

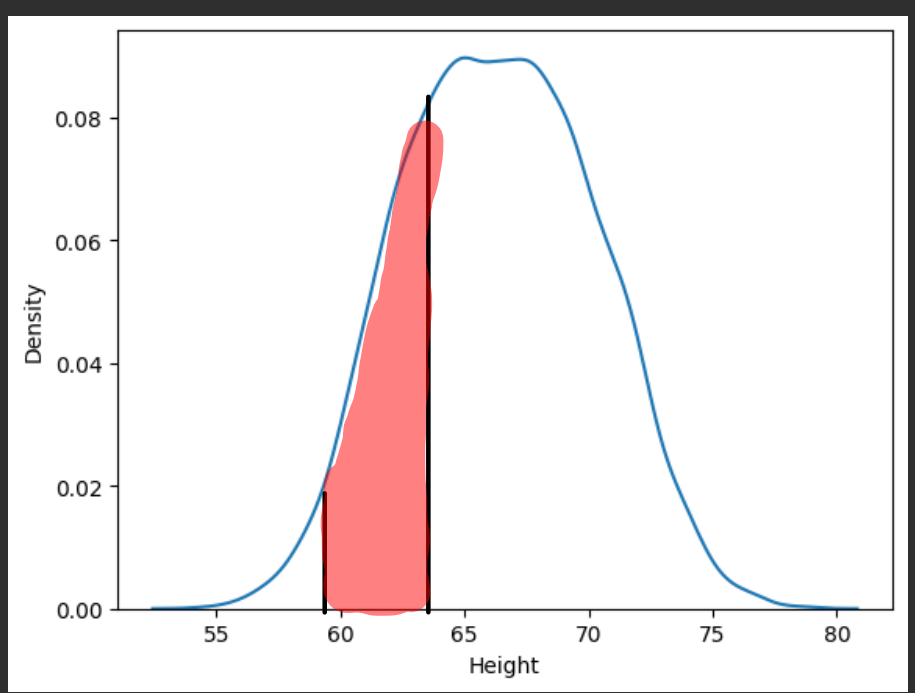




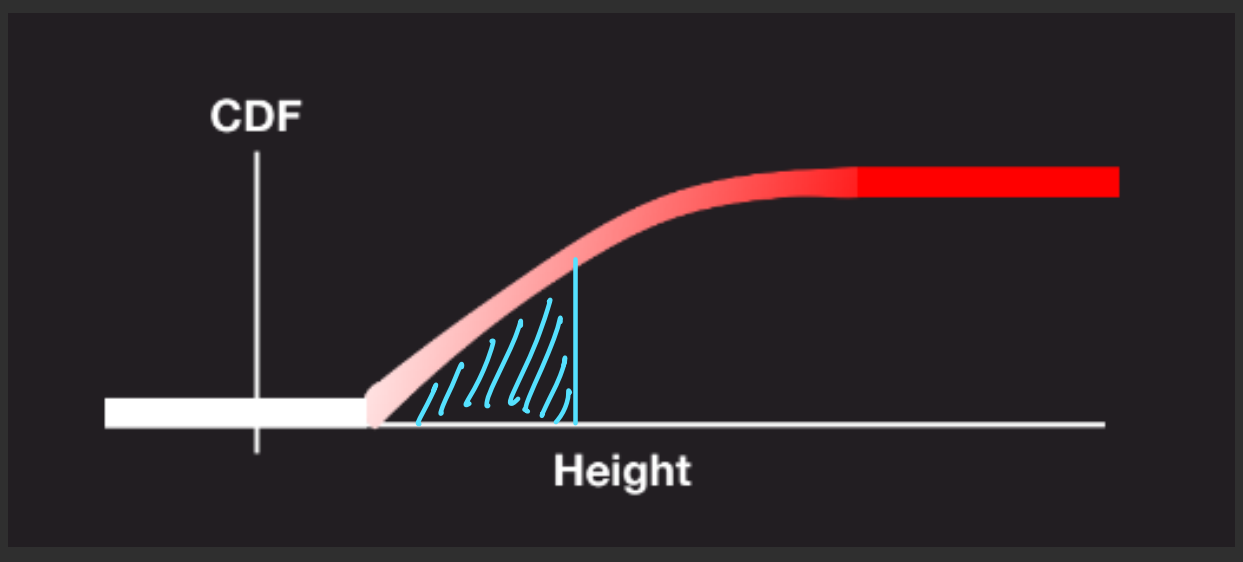
$$\left\{\frac{1}{6}+\frac{1}{6}+\frac{1}{6}+\frac{1}{6}+\frac{1}{6}+\frac{1}{6}\right\} \Rightarrow 1$$



(Discrete)



(Continous)



(Both)

Case Study (Empirical vs Theoretical)

P(H) = 0.5
P(T) = 0.5

A bag has 3 Red and 2 Blue balls.
You pick a ball, write its colour, and put it back in the bag. This is done 4 times in total.
If all 4 times, the Red balls was drawn, you win Rs 150.
Otherwise you lose Rs 10.
Question : Would engaging in this game result in a profit or loss for you?

3/5

$\begin{cases} P(X = \text{Red Ball}) = 3/5 \\ P(X = \text{Blue Ball}) = 2/5 \end{cases}$

R R R R ⇒ 150

I am putting the ball back hay

−10

10 times → [7 H] [3 T]

7/10

10,000 times → 5000 Heads & 5000 tails

simulation ↳ time lagti

3R
2B

P(R) = 3/5
P(B) = 2/5

→ [R R R B]
→ 3/5 × 3/5 × 3/5 × 2/5

[R R B B] ✓
3/5 × 3/5 × 2/5 × 2/5

R B R B R B B R B R R B
B R B R B B R R B R B R

P(X=1) = ${}^4C_1 \times (2/5)^3 \times (3/5)^1$

$= {}^4C_1 \times (3/5)^1 \times (2/5)^3$

$P(X=2) = {}^4C_2 \times (3/5)^2 \times (2/5)^2$

$P(X=3) = {}^4C_3 \times (3/5)^3 \times (2/5)^1$

Binomial = $\boxed{{}^nC_k \times P(s)^k \times P(f)^{n-k}}$

n → Total no. of trials

k → No. of successful event

P(s) → probability of success

P(f) → 1 − P(s) ⇒ failure

$n = 10$

$p = 1/4$

$P(X \geq 4)$

$P(X \leq 2)$

$P(X <$

$1 - P(X \geq 2)$

$= 1 - [P(X=0) + P(X=1) + P(X=2) + P(X=3)]$

$= 1 - [{}^{10}C_0 (1/4)^0 \times (3/4)^{10} + {}^{10}C_1 (1/4)^1 \times (3/4)^9 +$

$P(X=2) = {}^{10}C_2 \times (1/4)^2 \times (3/4)^8$

Empirical Approach

↳ Experimental ✓

1) Fair Coin

$P(H) = 0.5$

↳ 10 times ⟶ 7 Head 3 tails

$P(H) = 0.7$

2) 10,000 times ⟶ approx 5000 heads

Expectation using Empirical Approach

$$E(X) = \sum_{i=1}^{n} x_i \times P(x_i)$$

$E(X)$ → Random Variable

$\sum_{i=1}^{n} x_i \times P(x_i)$ → Weighted Average

Theoritical Approach

3R

2B

R → Red ball

B → Blue ball

$$\begin{cases} P(R) = 3/5 \\ P(B) = 2/5 \end{cases}$$

$$P(R) \times P(B)$$

↳ $3/5 \times 2/5$

[R, R, R, B]

↳ $(3/5) \times (3/5) \times (3/5) \times (2/5)$

↳ Order not important

[R, R, B, R] = [R, R, R, B]

R R R B ⇒ $(3/5) \times (3/5) \times (3/5) \times (2/5)$ = ✓

R R B B ⇒ $(3/5) \times (3/5) \times (2/5) \times (2/5)$ = ✓

$X = 1$ [No of red balls]

C_1 BBBR ⟶ $P(X=1) = (2/5)^3 \times (3/5)^1$

C_2 BBRB ⟶ $P(X=1) = (2/5)^3 \times (3/5)^1$

C_3 BRBB ⟶

C_4 RBBB ⟶

Same

$^4C_1 = \frac{4!}{1! \times 3!}$

$P(X=1) = C_1$ or C_2 or C_3 or C_4

$\Rightarrow \frac{4 \times 3!}{3!} = 4$

$\Rightarrow 4 \times \left[(2/5)^3 \times (3/5)^1 \right]$

$4C_1$ ↓ $P(B)$ $P(R)$

$$P[X=1] = 4\left\{\left(\frac{2}{5}\right)^3 \times \left(\frac{3}{5}\right)^1\right\}$$

$$P[X=2] = 6\left\{\left(\frac{2}{5}\right)^2 \times \left(\frac{3}{5}\right)^2\right\}$$

R R B B
R B R B
.
.
.
.

6 cases $\rightarrow$ ${}^4C_2$

$$P[X = k] = {}^{n}C_{k} \times \left\{\frac{3}{5}\right\}^{k} \times \left\{\frac{2}{5}\right\}^{n-k}$$

No of red balls

$$P[X = 1] = {}^{4}C_{1} \times \left\{\frac{3}{5}\right\}^{1} \times \left\{\frac{2}{5}\right\}^{3}$$

Binomial distribution

$$P(X = k) = {}^{n}C_{k} \times [P(S)]^{k} \times [P(F)]^{n-k}$$

$n \longrightarrow$ No of trials

$k \longrightarrow$ No of red balls / No of sucessful event [4R]

$P(S) \longrightarrow$ Prob of getting 4 red balls

$P(F) \longrightarrow 1 - P(S)$

Binomial dist

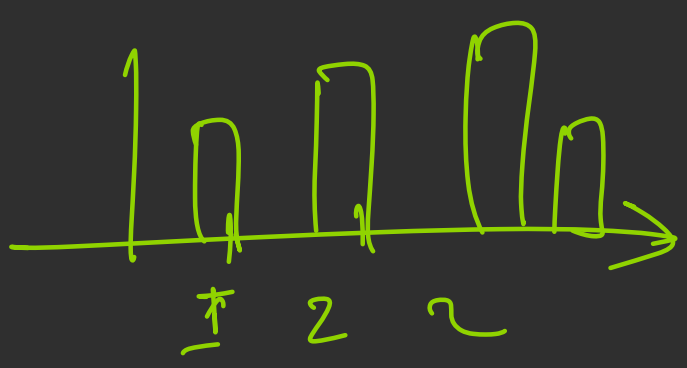
T F

dataset

Rain fall

Yes/No

Binomial



days of the week	Rainfal
0 Tues	Yes/No
1 Wed	Yes/No
2 Thus	Yes/No
3 Fri.	
:	